Cue card · Weather & durian: free farm apps for Agriculture

Woo Brothers · Jerantut · 20 minutes · planned 19 min, 1 min for questions

> **Option A · Windows laptop:** projector cable in → **Win + P** → **Extend** → open the talk in Chrome/Edge → drag it to the projector → **P** Projector Mode → **S** presenter window on the laptop (notes, next slide, pace timer). Next: → / Space / click / clicker · Back: ← · Skip screenshots: **Shift +** → · Black screen: **B**.

> **Option B · Phone only:** open the talk online → tap **Save offline** → set the language → HDMI adapter or screen mirroring → turn the phone sideways → ▶ **Present**. Tap right = next · tap left = back · tap the top edge = control bar (Exit, Skip ⏭, language, All slides, ■ Black, time left). Android buzzes when you are 30 s behind. The audience sees the whole phone screen, so use this card for notes.

1	0:00	**Title** 0.5 min	Selamat pagi / 大家好. I'm Stanley Woo. My friend Bryan Woo and I share a surname, so we call ourselves Woo Brothers. In 20 minutes: a weather forecast you can trust for your own farm, a set of durian apps in the order you'd use them in a season, and how we use Claude AI to build all of this. Everything is free.
2	0:30	**Follow on your phone** 0.25 min	Invite everyone to scan now: the whole talk opens on their phone, with every QR code and screenshot, and they can forward it on WhatsApp to friends who couldn't come. If someone is slow, the same QR code is on the last slide.
3	0:45	**woobrothers.my** 1.25 min	Show woobrothers.my first: it is the home for everything today, so if people remember one thing, it's this address. We are not a company and sell nothing, and new apps keep arriving, so we don't quote a number. Bryan and I are good friends who build these together. Every app has BM, English and 中文. Invite them to scan now and add it to the home screen; every app in this talk is linked from there.
4	2:00	**WeatherNext Jerantut** 1.25 min	Our app built for this district. Three local spots come preloaded (Kg. Simpang Tebing Tinggi, Damak, Bukit Seladang). Walk through the cards; stress the Phytophthora score for durian and the WhatsApp broadcast, which lists farms north to south. Be honest: disease scores are calibrated estimates, not yet field-validated, so we want Jerantut growers to tell us when they're wrong. The AI briefing is optional (the user's own free Gemini key); forecast and disease risk work without it.
5	3:15	Screenshots: 而连突农友天气 · WeatherNext *(15 s · optional, tap Skip ⏭)*	

6	3:30	**Install** 0.5 min	Stop talking and let people scan. iPhone users must open the link in Safari to get "Add to Home Screen". Offer to help anyone after the talk.
7	4:00	**Why forecasts differ** 0.75 min	Bridge to CompareCast. Three steps: measure, compute, and every centre runs its own model. In the tropics, a model grid box is about 9 km but an afternoon thunderstorm cell is only a few km, so a model can be right about the storm and still put it over the next kampung. Treat "% chance of rain" as a chance.
8	4:45	**CompareCast + live chart** 1.5 min	This chart pulls live data for Jerantut town when the phone or laptop is online; offline it shows the run saved on 11 Nov 2026 and the label says so. Point at the highlighted day: same town, same day, very different numbers. Ask: "Which one would you trust?" Answer: whichever has been right at your farm. CompareCast shows forecast against actual rain, a gust map, MetMalaysia radar and per-plot rain reminders. If the chart is slow, press Refresh or move on.
9	6:15	Screenshots: CompareCast *(15 s · optional, tap Skip ⏭)*	
10	6:30	**RainLog Lab** 0.5 min	RainLog Lab looks backwards: rainfall history for each orchard, the current dry streak and the longest dry spell this year, which is what durian growers watch before flowering. You set what counts as a rainy day in mm. 114 orchards come preloaded; you can add your own by GPS, export to Excel, share an image.
11	7:00	Screenshots: RainLog Lab *(15 s · optional, tap Skip ⏭)*	
12	7:15	**RainScore Lab** 0.5 min	RainScore Lab scores six models (IFS 9 km, ICON, JMA, CMA, GEM, GFS) against ERA5 actual rain, every day, for 114 orchards. Two scores: average miss in mm, and how often it called wet or dry correctly, over 7, 14 or 30 days. This closes the loop from the CompareCast slide: now you know which line on that chart to believe.
13	7:45	Screenshots: RainScore Lab *(15 s · optional, tap Skip ⏭)*	
14	8:00	**ENSO Monitor** 0.5 min	From days to months. ENSO Monitor reads NOAA's official RONI index and weekly Niño 3.4 sea temperature, then adds the Indian Ocean Dipole, the MJO (a wet or dry pulse every 30–60 days) and Northeast Monsoon cold surges. We are in the Northeast Monsoon now (November), so drains and Phytophthora come first. Each signal comes with orchard advice, and it links to MetMalaysia for official warnings. SUHU is the sister app for drought and haze risk.
15	8:30	Screenshots: ENSO Monitor *(15 s · optional, tap Skip ⏭)*	

16	8:45	**NDVI Remote Sensing** 0.5 min	The NDVI app uses free Sentinel-2 satellite images (10 m pixels, a new pass every few days) to compute NDVI, a plant-health index, and NDMI, a moisture index, then marks the weaker zones of each plot and shows local weather. Instead of walking every row, go straight to the patch that looks different. Be honest about the tropics: clouds often block the view, so it uses the most recent clear image. This is also the bridge into the durian section: from the sky down to the tree.
17	9:15	Screenshots: NDVI Remote Sensing *(15 s · optional, tap Skip ⏭)*	
18	9:30	**Durian Tree Science** 0.5 min	The durian section follows the order you'd meet things in a season: know the tree, protect it, keep the soil alive, time the season, fix fruit quality, feed it, then spray. Durian Tree Science is the foundation: 45 sections and 31 plates, from roots and trunk to floral induction, fruit set and fruit drop, plus how Thailand, Vietnam, Indonesia and others grow durian.
19	10:00	Screenshots: Durian Tree Science *(15 s · optional, tap Skip ⏭)*	
20	10:15	**Durian IPM** 0.5 min	Durian IPM: tap where the problem is and what you see; it places it in one of about 22 pest groups. Non-chemical steps come first, then organic and conventional options ranked by effect. It suggests a rotation from your last spray so you don't repeat a mode of action, and warns about bees at flowering, harvest intervals and banned products.
21	10:45	Screenshots: Durian IPM *(15 s · optional, tap Skip ⏭)*	
22	11:00	**AMIS · AgriMicrobe** 0.5 min	AMIS covers 83 beneficial microbes and 42 crops. The Synergy Guide shows which microbes help or fight each other; the Tank-Mix Checker says whether microbes, foliar fertilisers and pesticides can share a tank; First Aid gives today's actions by urgency, including when microbes are not the answer. Most of it works offline; the AI advisor needs the user's own API key.
23	11:30	Screenshots: AMIS · AgriMicrobe *(15 s · optional, tap Skip ⏭)*	

24	11:45	**AgriPro Timeline** 0.75 min	AgriPro Timeline, Personal Edition. The backbone is days after anthesis (days from full bloom), which is how growers and researchers time durian harvest: about 95 days for Musang King, 105 for IOI, around 120 for D24 and Black Thorn. You anchor a bud, bloom or harvest date and it lays out all 7 stages. Each time you log what you saw on the tree, the projection re-sets to your observation. A heat-unit (GDD) adjustment nudges dates for hot or cool spells; its base temperature is a working assumption to refine against your own records. Then send the schedule to WhatsApp.
25	12:30	Screenshots: AgriPro Timeline *(15 s · optional, tap Skip ⏭)*	
26	12:45	**Offseason Durian** 0.5 min	Off-season: 8 steps from choosing the harvest month, preparing the tree, PBZ, drying the root zone, buds to bloom, holding the crop, harvest, and resetting so PBZ doesn't build up. Off-season fruit can fetch 2–3 times the peak price, but it is hard; every figure here links to a published trial.
27	13:15	Screenshots: Offseason Durian *(15 s · optional, tap Skip ⏭)*	
28	13:30	**Tiger Stripe** 0.5 min	Tiger stripe (老虎斑, belang harimau) is a big quality problem for Musang King. The app turns Prof. Dr. Tran Van Hau's AGRI Malaysia 2026 talk into notes and an action plan: affected trees had low leaf potassium, calcium and magnesium and high nitrogen, consistent with calcium and carbohydrate shortage. The action plan converts the paper's calcium nitrate and magnesium sulphate spray rates to the farm-shop product you actually bought. It leads into the nutrient notebook on the next slide.
29	14:00	Screenshots: Tiger Stripe 老虎斑 *(15 s · optional, tap Skip ⏭)*	
30	14:15	**AgriPro Notebook** 0.5 min	The nutrient notebook: tap any nutrient to see its job and which nutrients it blocks or helps. The pH chart shows what a root can take up at each soil pH. Mixing rules explain how to avoid gelling, and the advisor suggests a recipe by crop and durian variety.
31	14:45	Screenshots: AgriPro Notebook *(15 s · optional, tap Skip ⏭)*	
32	15:00	**Spray day: ChemLog + DroneDose** 0.75 min	Spray day. ChemLog Lab 2: enter tank volume and rates, get per-tank amounts and pack counts, send to workers on WhatsApp, and log every round by farm, zone and worker with stock and cost. DroneDose: pick the DJI model and tank, enter mapped area and rate, and it works out loads, water and product per tank, then copies a mixing sheet for the pilot. Both work offline.
33	15:45	Screenshots: ChemLog Lab 2 + DroneDose *(15 s · optional, tap Skip ⏭)*	

34	16:00	**More durian apps** 0.25 min	Pick one or two depending on the room. Borer Action and Trunk Disease Triage are the most asked-about in durian areas: tell pinhole borer from longhorn and Phytophthora before spending money. Nutrient Triage is a leaf key. TerraDurian uses NASA satellite soil moisture for the flowering signal. DurianExport covers documents and flights for selling to China (not an official SF Express app). Tap a card to show its QR.
35	16:15	**Claude: how we build** 1.25 min	How can two people with day jobs build so many apps? We use Claude, an AI assistant from Anthropic, as our coding partner. We bring the farm problem, the research and the field checking; Claude writes and tests the code and helps with the three languages. Step 4 matters most: nothing goes out until we've checked it against sources and real trees. Then mention this deck was built with Claude too. (Adjust this slide if your own workflow differs.)
36	17:30	**Claude: try it** 1 min	Read one example aloud in the audience's language. Claude has a free plan on the web and as an Android/iPhone app; you can type in BM or Chinese and attach a photo of a leaf or a label. Then the four rules: AI can be wrong, so check the tree and ask an agronomist; MetMalaysia for official warnings; follow the pesticide label; never type passwords, IC or bank details.
37	18:30	**Thank you** 0.5 min	Thank the organisers and growers. Take questions. Repeat the ask: tell us when a forecast or a diagnosis is wrong; that is how these apps get better for Jerantut. Leave this slide up so people can scan or copy the WhatsApp numbers.